Building a financial data pipeline with Kafka

Legalese…

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AHL at a glance

Man AHL

100% systematic quant

$19bn+ AUM

Everyone writes code

65+ Tech

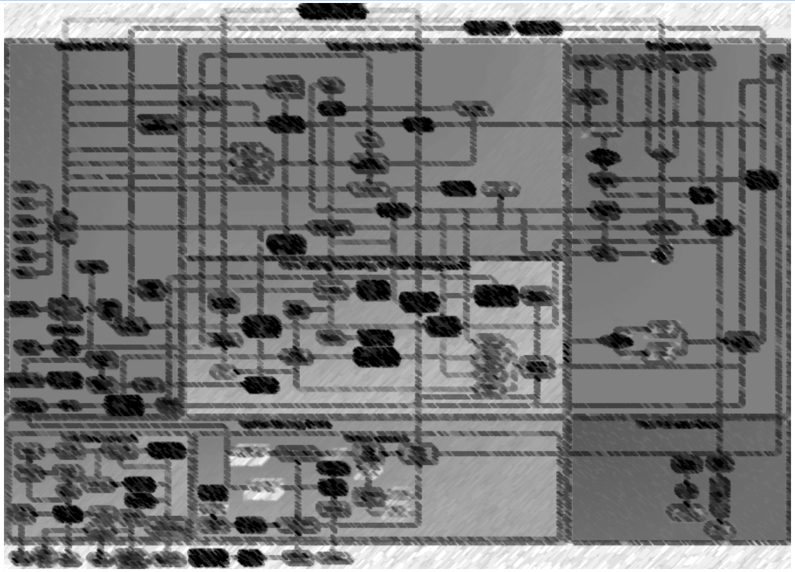
30 years experience

Open culture

Open source software

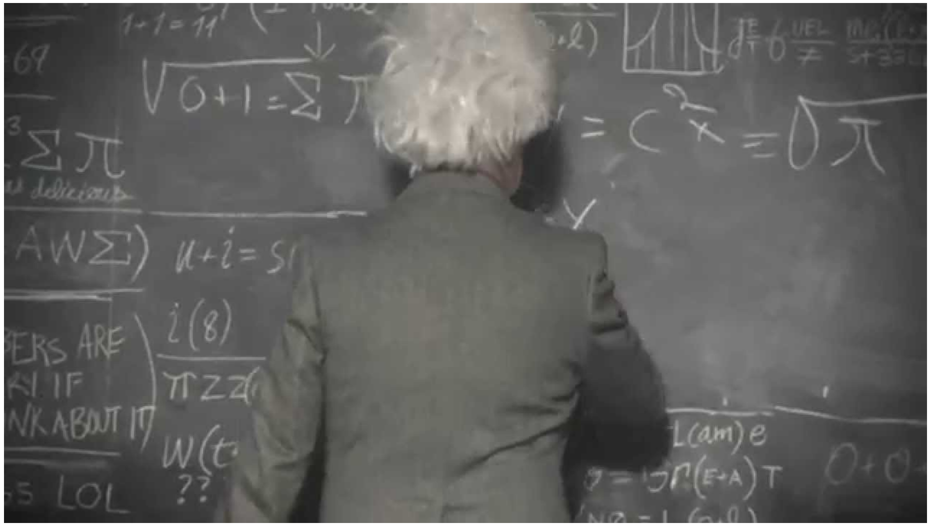
Apache kafka Apache Spark theano

arctic PyBlogs PyData London



25 **Core** technologists

- Linux infrastructure
- Data engineering
- Core platform
- Core trading
- Electronic execution
- Risk systems



35 **Research** technologists

- Across 10 research teams
- Support research
- Implement strategies
- Maintain strategies

The problem

The AHL streaming data pipeline

From the sky

Something happens somewhere on 

The AHL streaming data pipeline

From the sky

Something happens somewhere on 

????

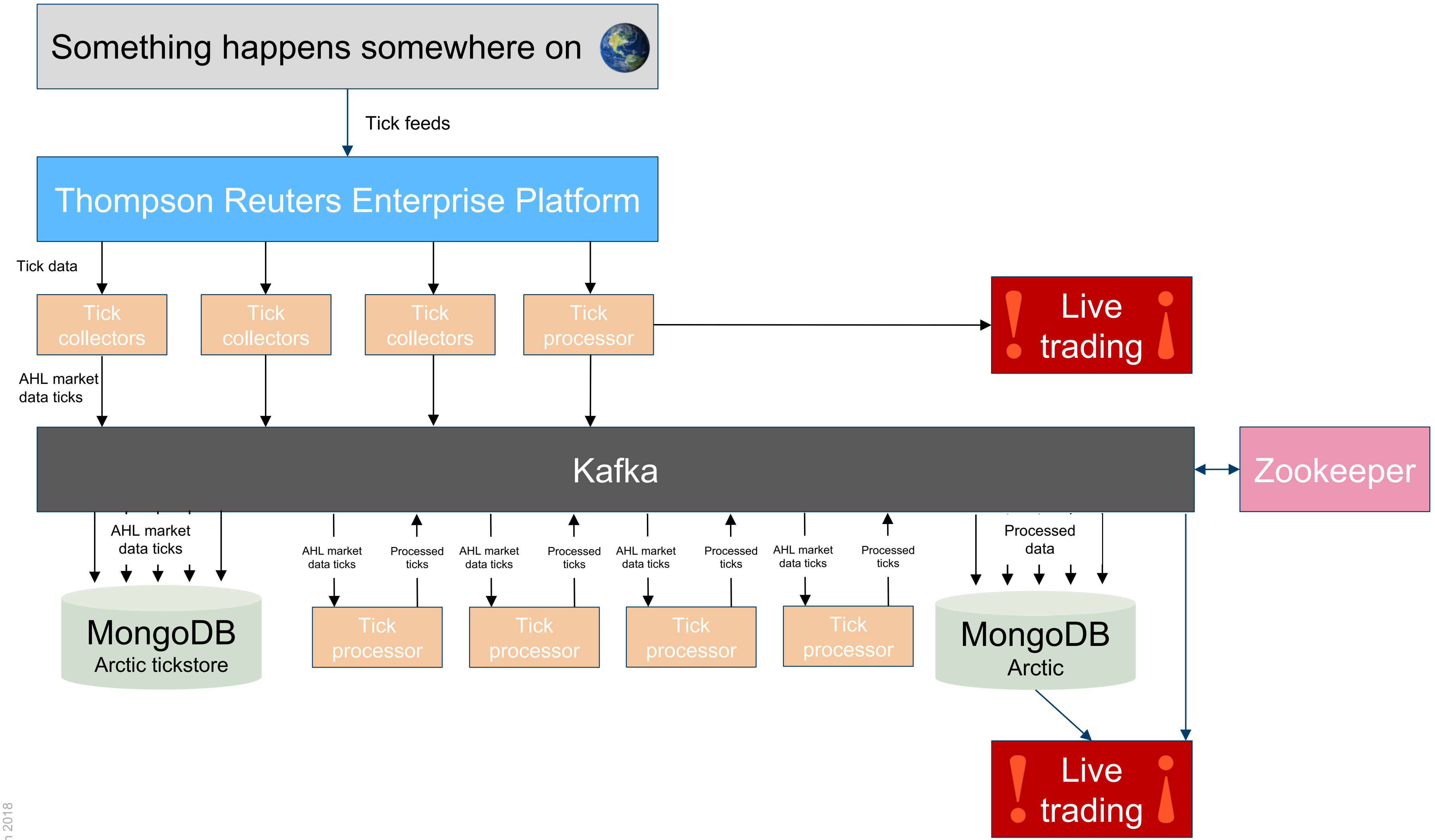
Data

Strategies

! Live trading !

The AHL streaming data pipeline

From the sky



The AHL streaming data pipeline

Something happens somewhere on 

Tick feeds

Thompson Reuters Enterprise Platform

Tick data

```
2018-02-07 15:59:18,712 INFO  [sub_worker_6] LoggingSubscriber - Type: UPDATE Subject: BPOD.TRI.US
                    PROD_PERM:         Long:                    14003
                   DelayedEID:         Long:                        0
                      UpdType:       String:                      ASK
                    ASK_MMID1:       String:                       UN
                       ASKXID:       String:                      NYS
                     ASK_TIME:    LocalTime:                    15:59
                    ASK_TIME1:    LocalTime:                 15:59:18
                   ASK_TIM_MS:         Long:                 57558635
                     AskPrice:       Double:                    40.87
                       AskLSc:       String:                       UN
                    LstAIsInd:         Long:                        0
                      AskTime:    LocalTime:                 15:59:18
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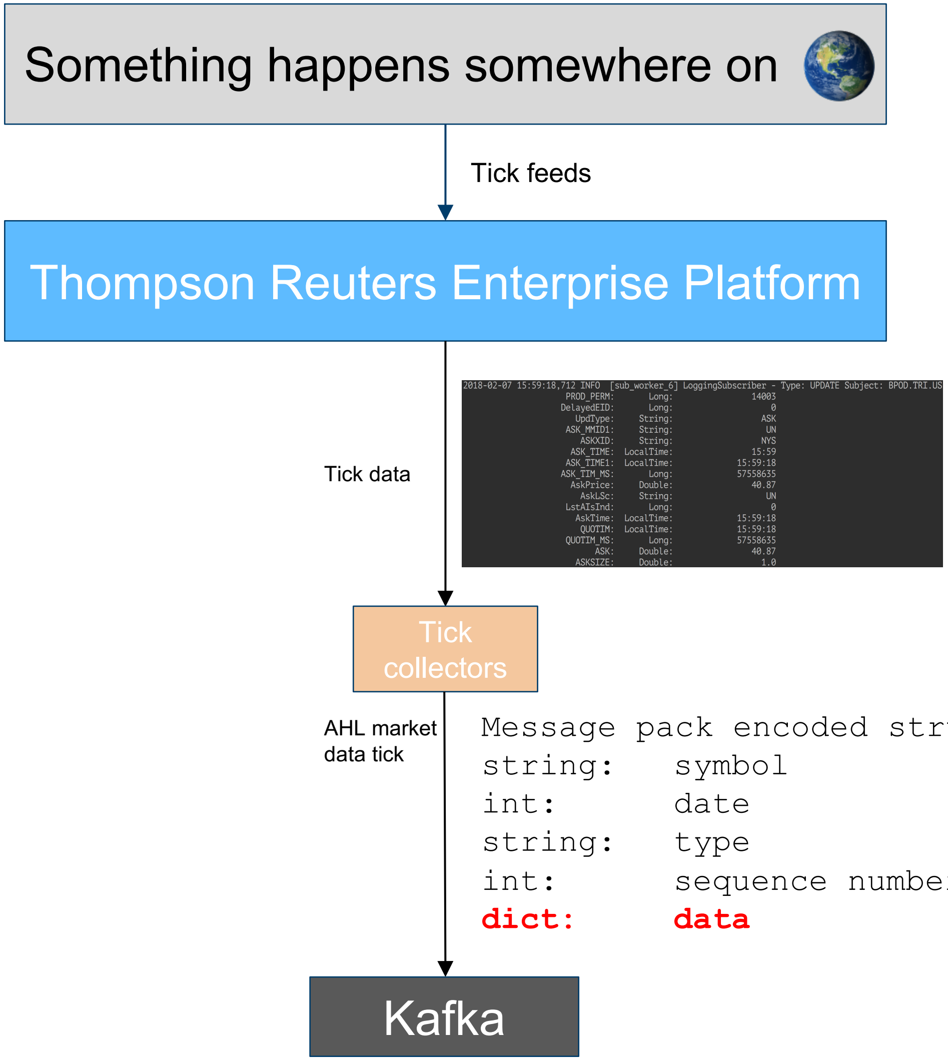
x2 billion a day

The AHL streaming data pipeline

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```
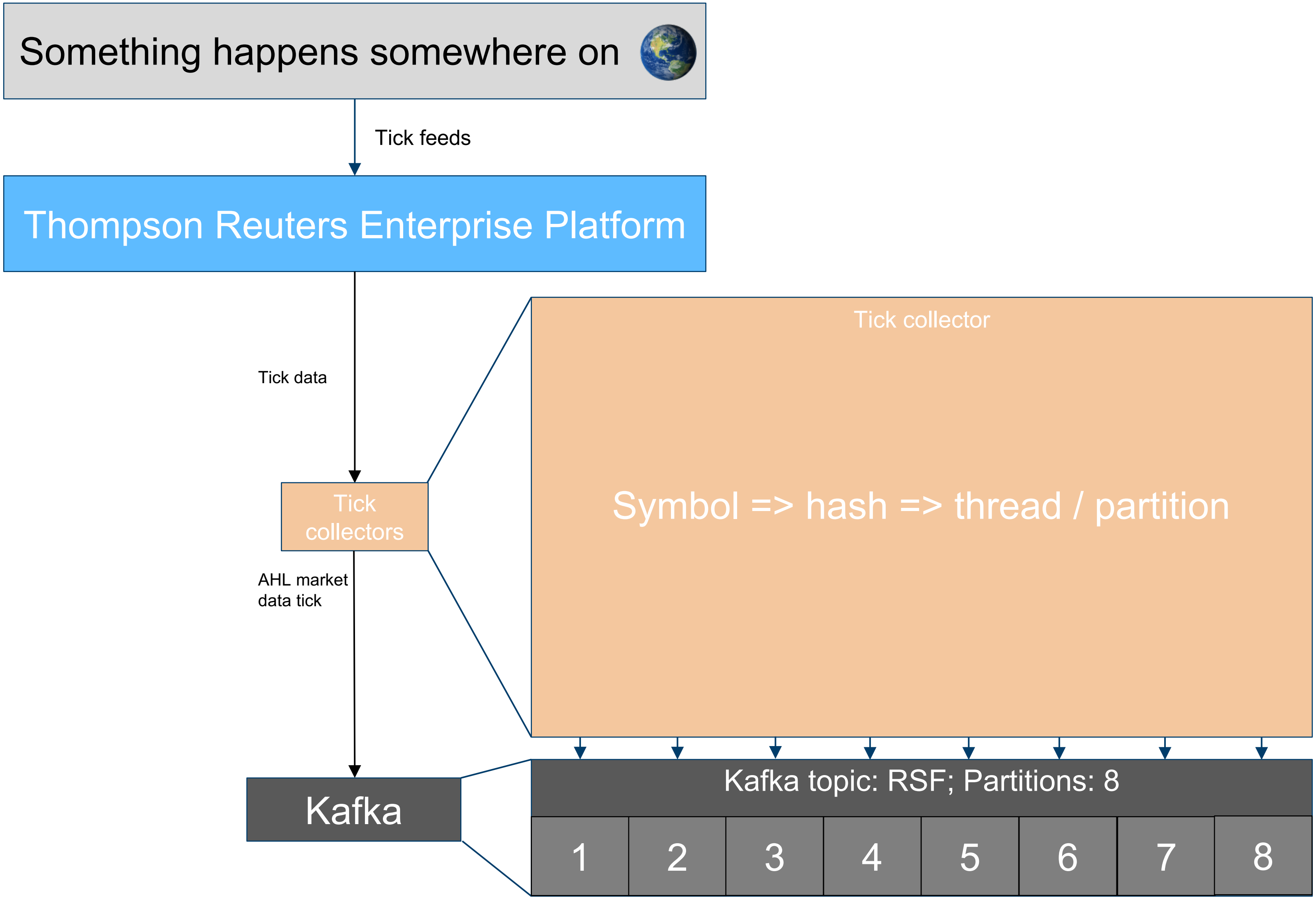
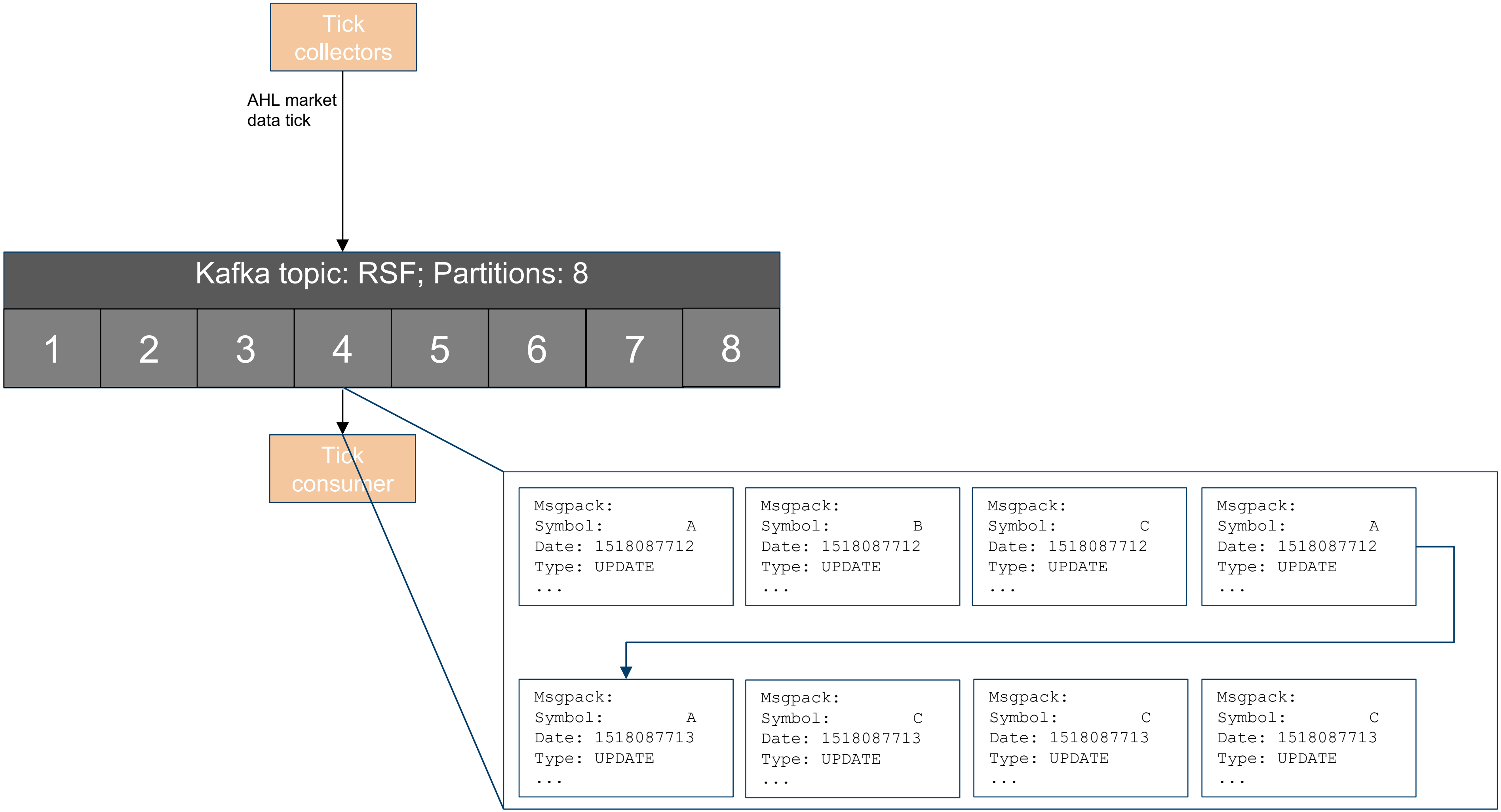

```
Message pack encoded structure:
string:   symbol
int:      date
string:   type
int:      sequence number
dict:     data
```

The AHL streaming data pipeline

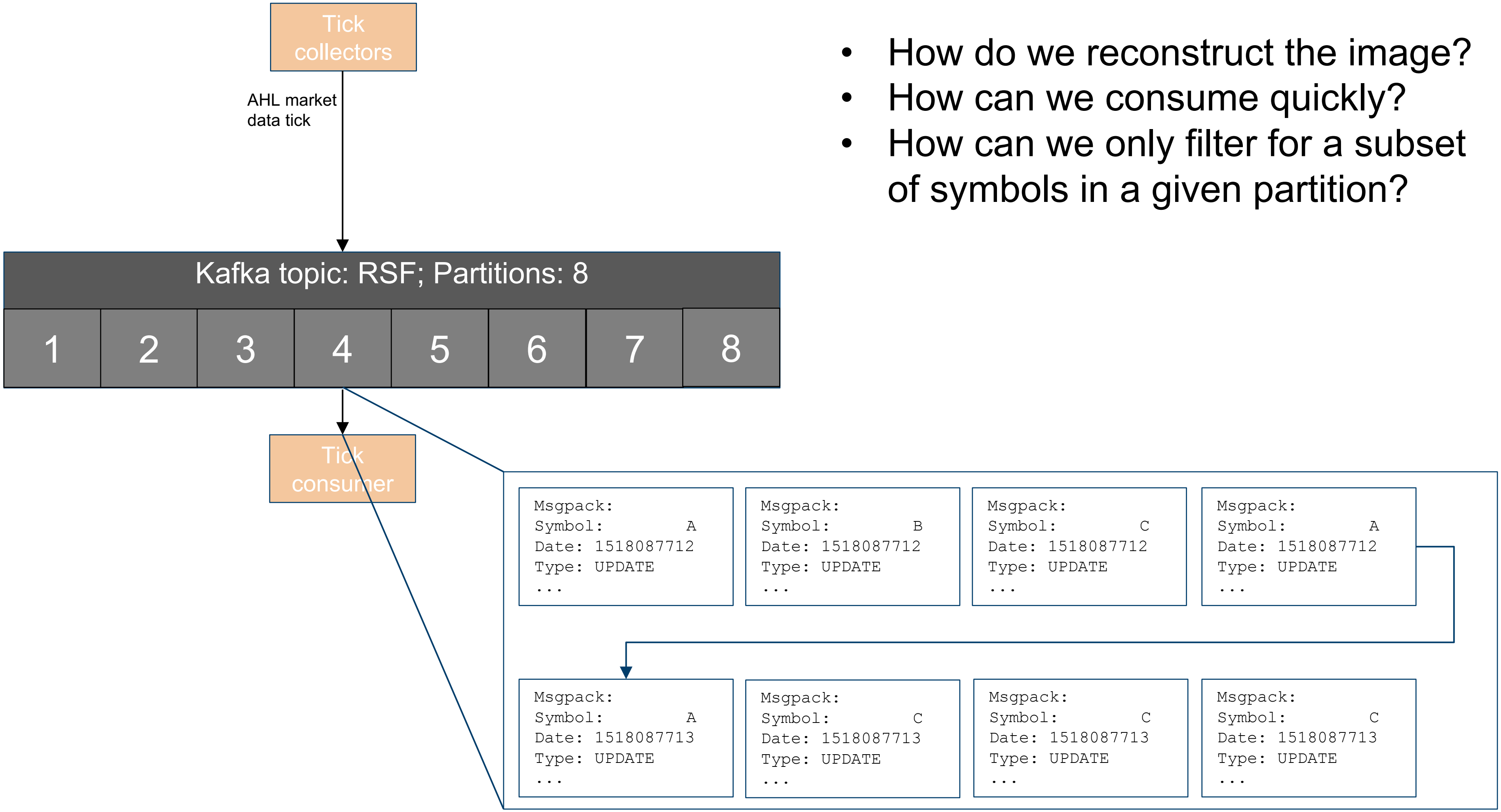


The AHL streaming data pipeline

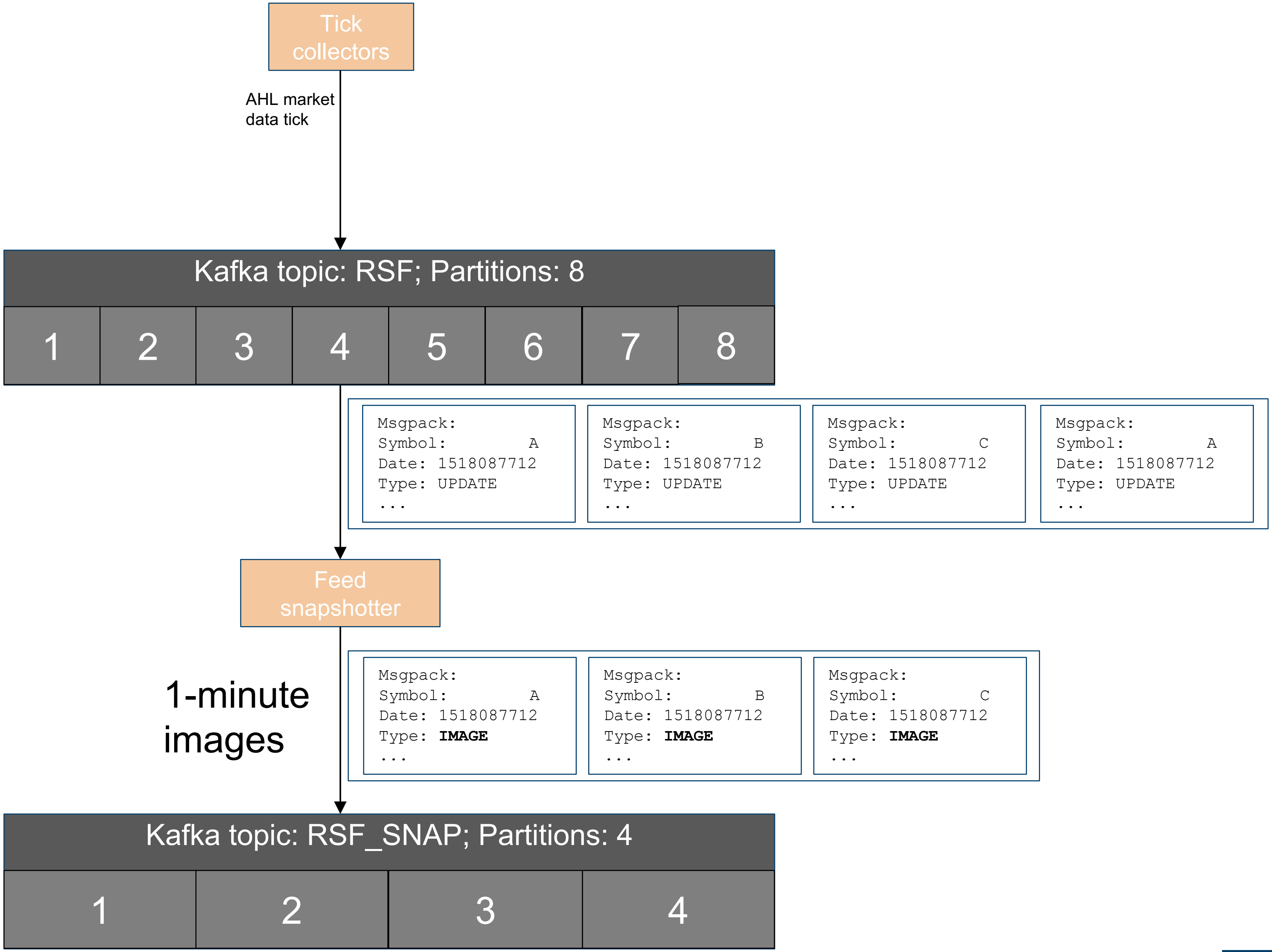


The AHL streaming data pipeline

- How do we reconstruct the image?
- How can we consume quickly?
- How can we only filter for a subset of symbols in a given partition?



The AHL streaming data pipeline



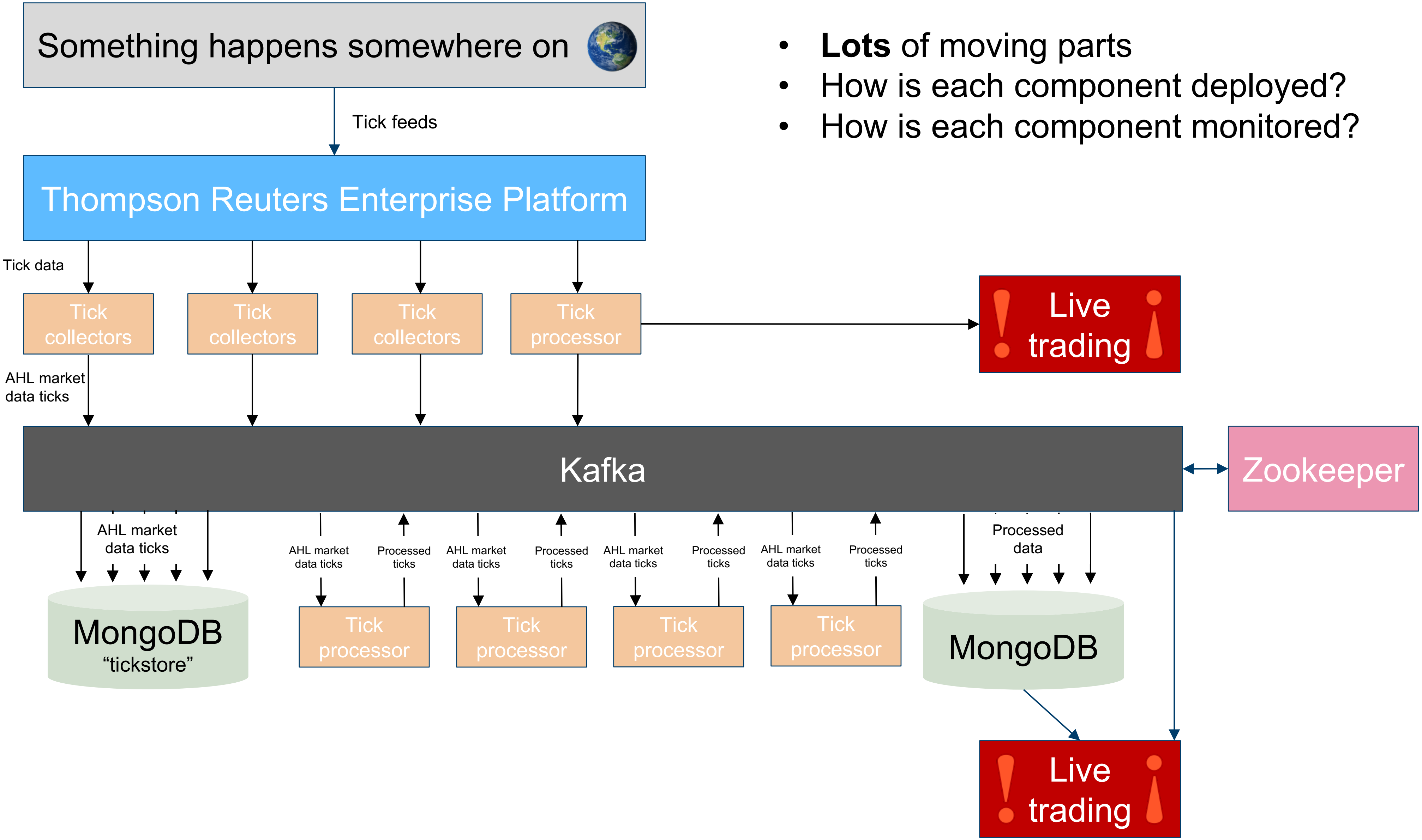
The AHL streaming data pipeline

AHL Kafka client

- AHL is a Python shop.
- Only one mature Python Kafka client in 2013 – kafka-python. Kafka-python is **pure Python** implementation. It’s also (relatively) slow to consume ☹
- We use a custom Kafka message set decoder written in Cython. ~10x speed up. Filtering based on Symbol done in Cython.
- Our Kafka clients (Java and Python) talk in terms of timestamps rather than Kafka offsets – does binary search into the partition behind the scenes.
- Time-based search now implemented natively in Kafka (since 0.10.1.0) on the broker side.
- Kafka-python is also slow to produce ☹ Python client creates a pool of producers. Maps from symbol -> producer -> partition.

The AHL streaming data pipeline

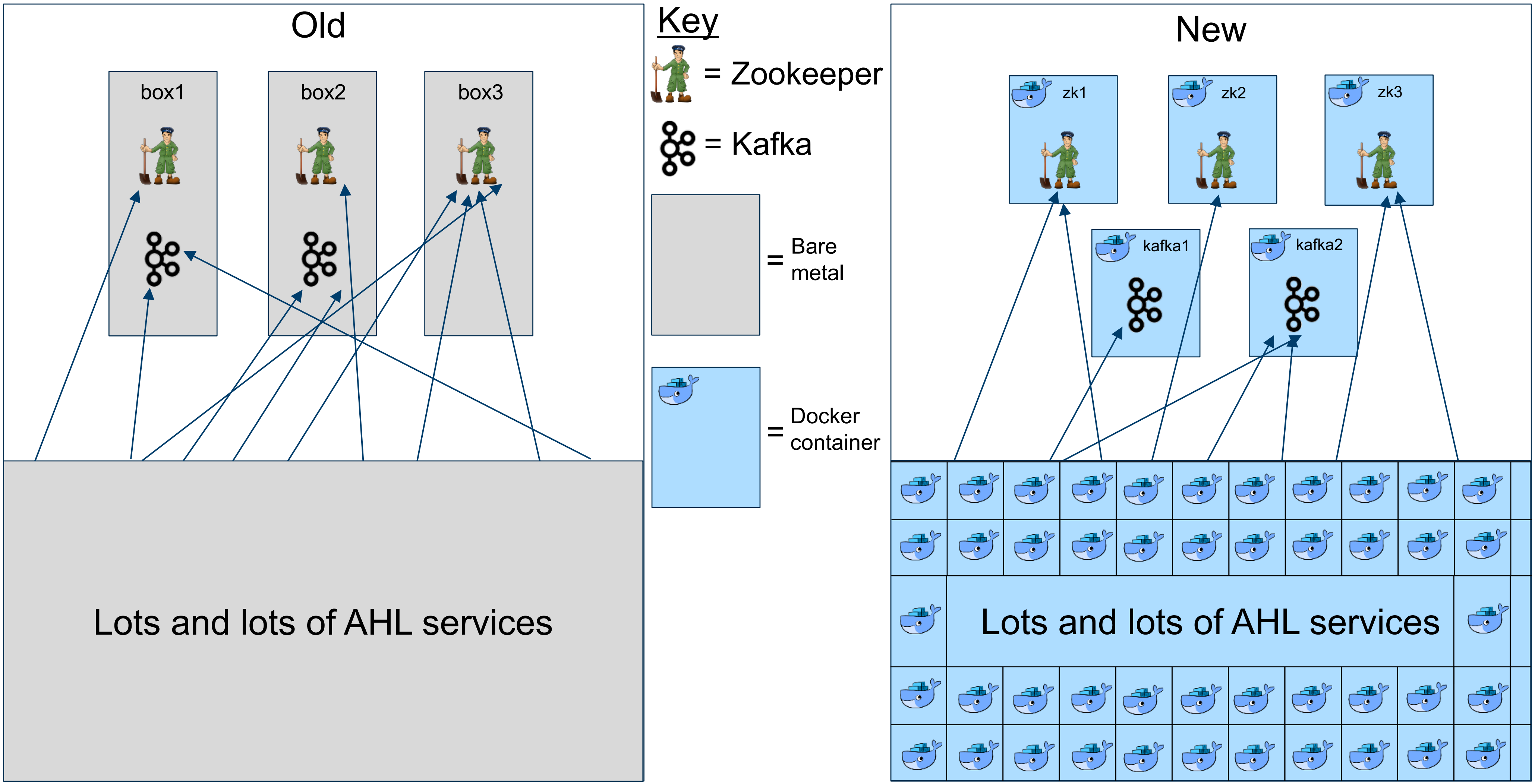
How is this deployed? How do we know it works?



- **Lots** of moving parts
- How is each component deployed?
- How is each component monitored?

The AHL streaming data pipeline

How is each component deployed?



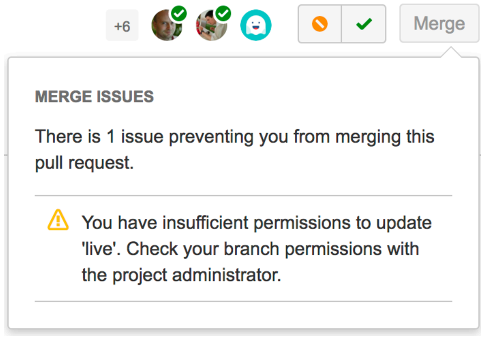
The AHL streaming data pipeline

How is each component deployed?

- At AHL we’ve drunk *some* of the Docker kool-aid.
- Dockerize everything!

- Docker-compose deploy everything!

- **No** Kubernetes, docker swarm, linkerd, etcd, etc.
  - We use manually managed CNAMEs heavily for aliasing/indirection

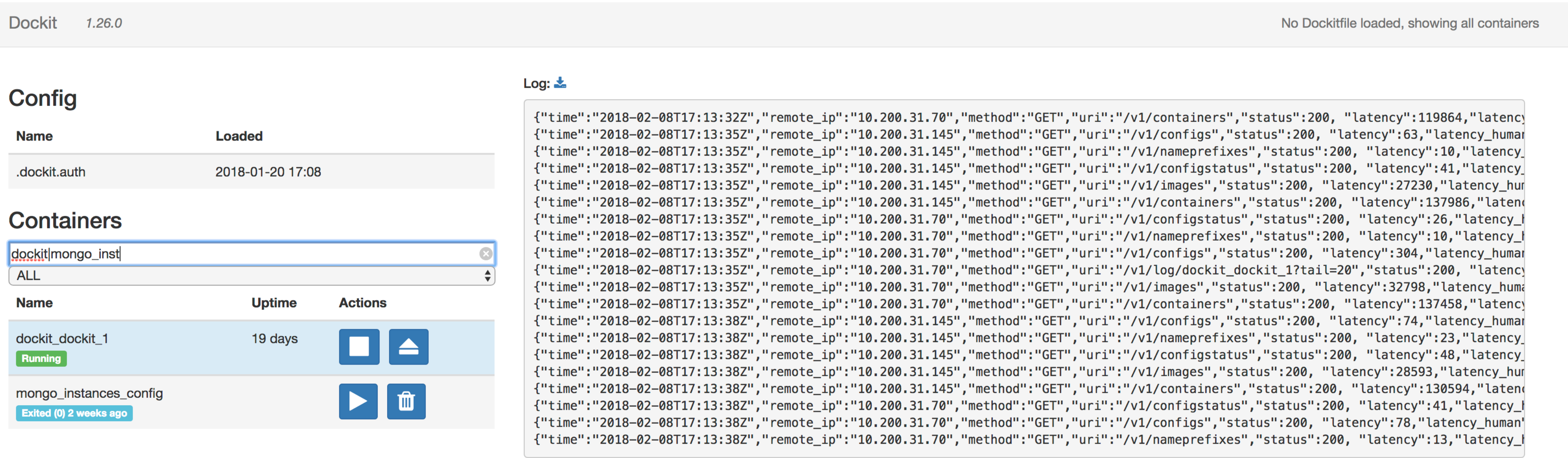


- We use our own internal solution for deploying docker-compose services
  - Heavily integrated with Bitbucket. Engineers and Researchers are **not allowed to touch the live trading system.**
  - Services are created and destroyed on merge. Merges are only permitted by support staff.
- For basic container administration, every box runs an instance of an agent called **Dockit.**

The AHL streaming data pipeline

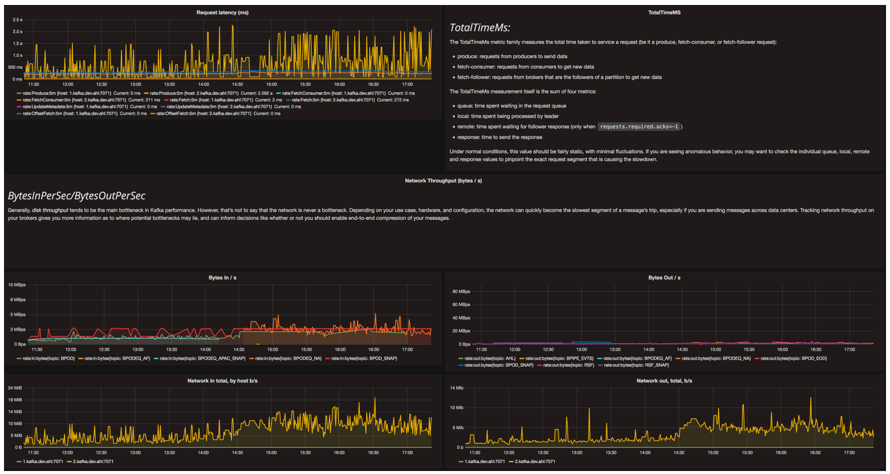
How is each component deployed?

- Dockit

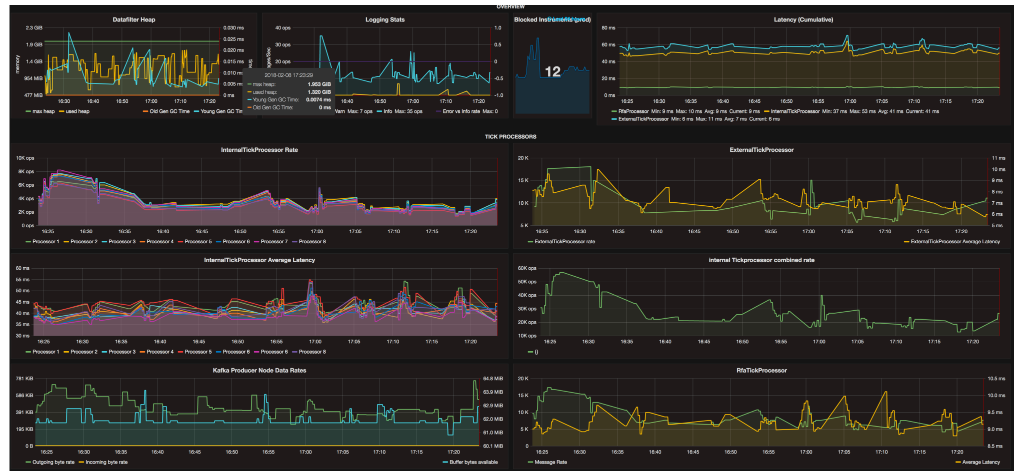


- We **need** to know when something goes wrong
- Prometheus + nagios + grafana + alerta

Kafka



Tickprocessor



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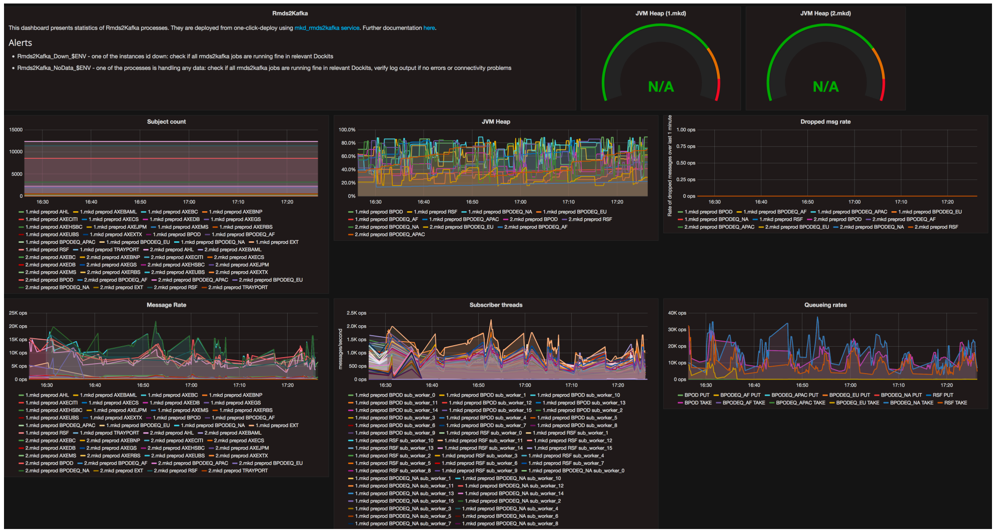
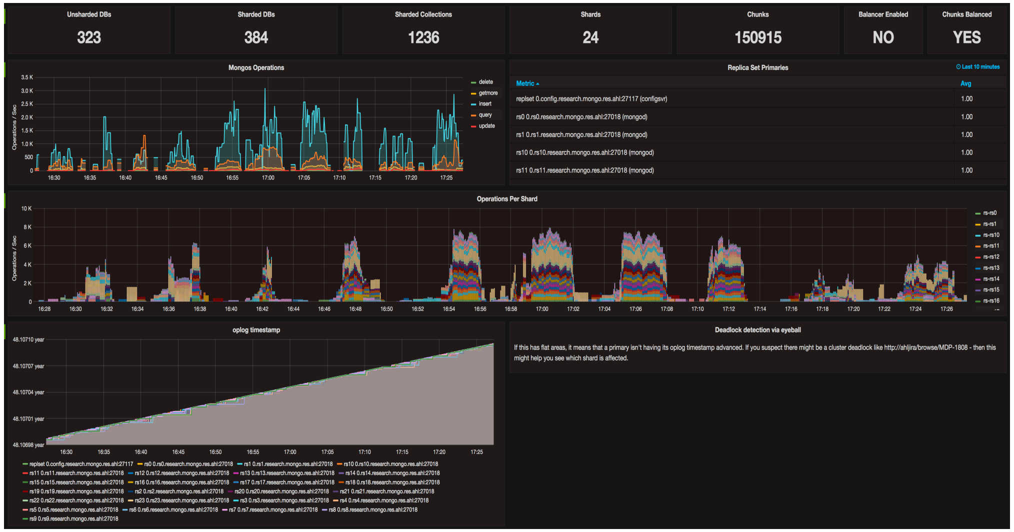
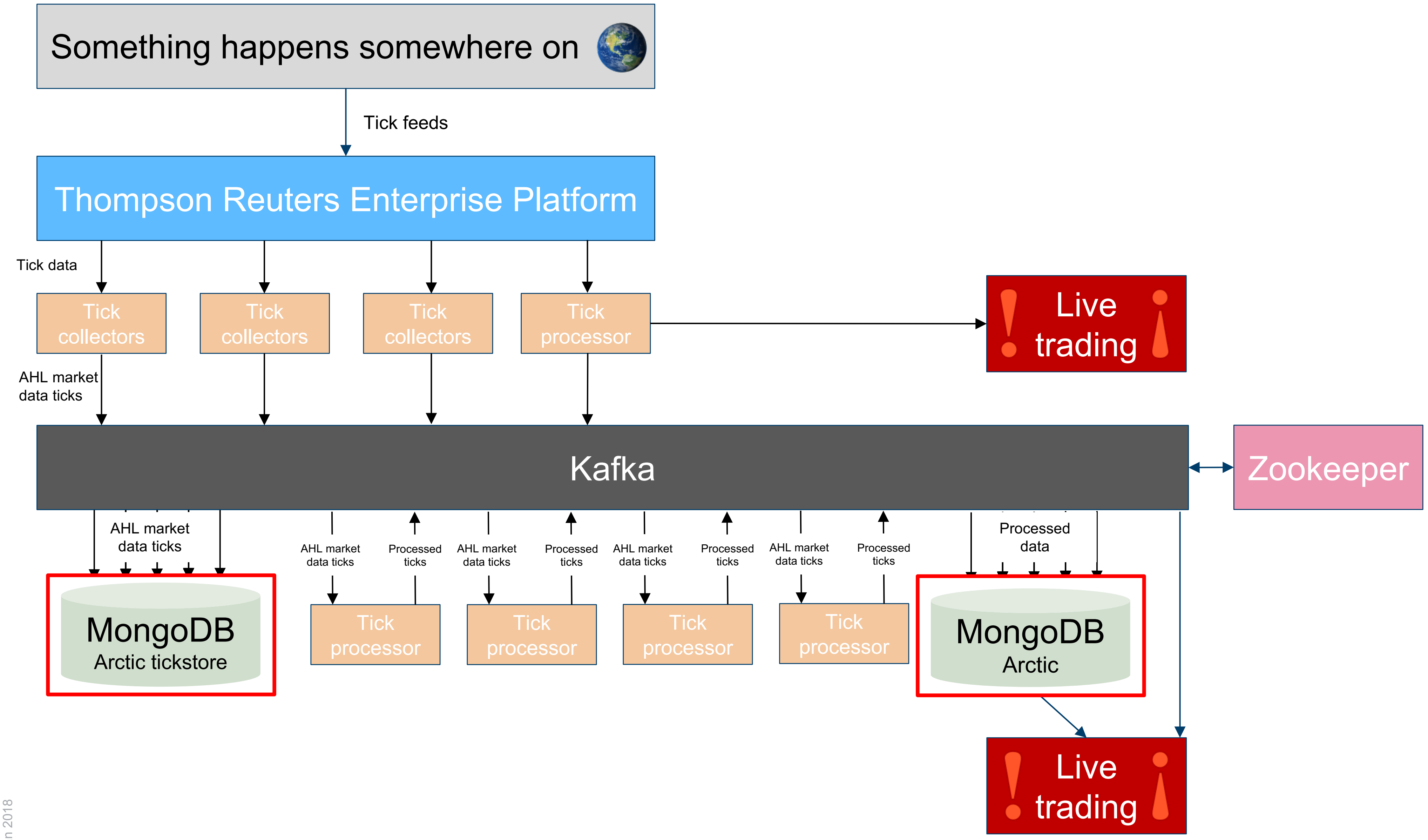
Tickcollector



Mongo



The AHL streaming data pipeline

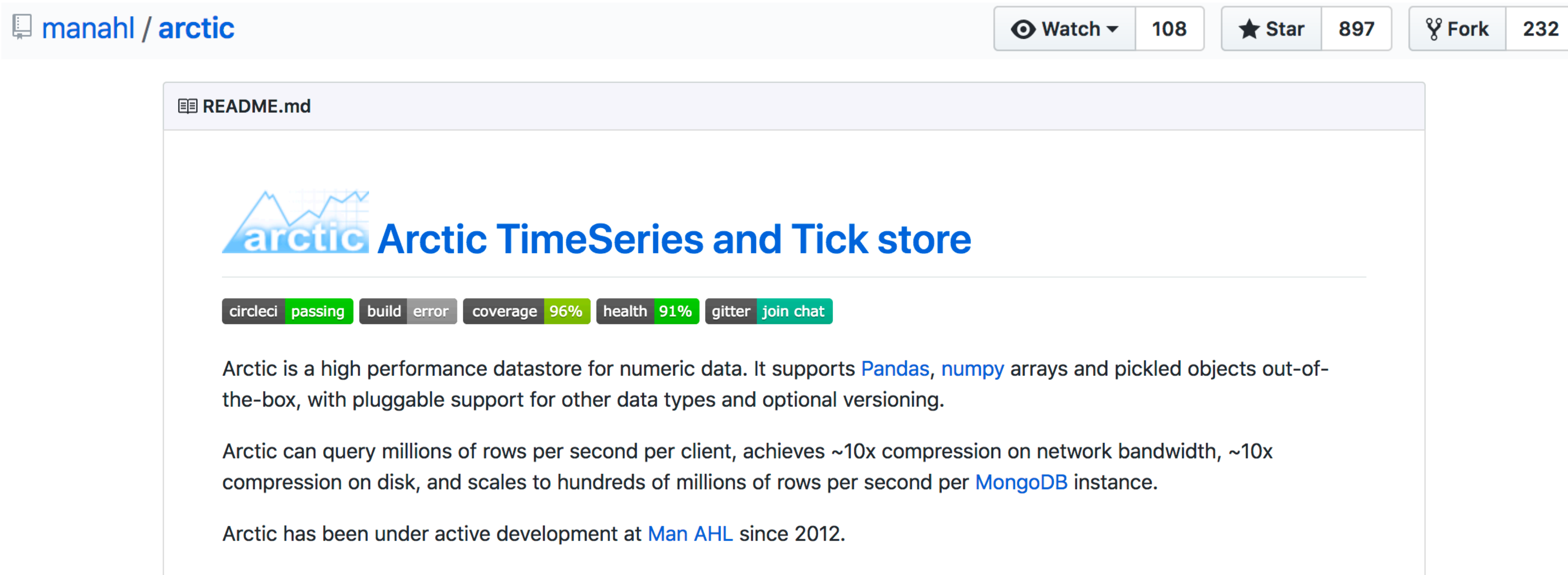


- Data sizes we deal with…
  - ~1MB x 1000s — 1x a day price data — 10k rows (30 years)
  - ~0.5GB x 1000s — 1-minute data — 4M rows (20 years)
  - ~1GB x 1000s — 10k x 10k data matrices — 100M cells (30 years)
  - ~30TB — Tick data — 200k msgs/s
    >1B msgs/day

- … Rand different shapes
  - Time series of prices
  - Event data
  - News data
  - Metadata
  - What's next?

- Requirements
  - Easy to use – *and we mean easy*
  - Fast – *as fast as local files*
  - Scalable – *unbounded in data-size and number of clients*
  - Agile – *any data shape; new shapes; iterative development*
  - Complete – *all data behind the simple API*

- We store all our data, of every type and shape, in Arctic. Arctic is an optionally versioned, columnar, time-series database backed by Mongo, which is entirely open source…



- Check it out at https://github.com/manahl/arctic
- Every tick, every EOD sample, every downsampled 1-minute-bar is stored permanently in Arctic.
- **Everything is retrievable - nothing is thrown away!**

Questions

Arctic is a high performance datastore for numeric data. It supports Pandas, numpy arrays and pickle objects out-of-the-box, with pluggable support for other data types and optional versioning.

Arctic can query millions of rows per second per client, achieves ~10x compression on network bandwidth, ~10x compression on disk, and scales to hundreds of millions of rows per second per MongoDB instance.

Arctic has been under active development at Man AHL since 2012.

@ManAHLTech

?

- We're hiring!

  mparrott@ahl.com